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Web Mining - Research Proposal

Exploring the Potential of LLMs in Opinion Mining

Synopsis

Opinion mining, also known as sentiment analysis, is a crucial task in natural language processing (NLP) that involves extracting and classifying subjective information from text, such as sentiments, emotions, and opinions [11]. Traditional sentiment analysis methods have primarily focused on binary sentiment classification (positive vs. negative), while more recent research has sought to address the need for fine-grained opinion mining. This research proposes the development of a novel approach to fine-grained opinion mining using pre-trained large language models (LLMs) to identify multiple aspects of private states in text, including:

- **Type:** The category of the private state (e.g., sentiment, emotion, intention).
- **Polarity:** The directional orientation of the opinion (positive, negative, neutral).
- **Intensity:** The strength or degree of the opinion (weak, strong).
- **Opinion expressions:** The specific words or phrases that convey the opinion.
- **Sources:** The individual or entity expressing the opinion.

This research will fine-tune both **pre-trained models**, such as BERT [1] and T5 [7], and **large language models (LLMs)** to improve performance across multiple dimensions of opinion mining. By comparing the effectiveness of traditional pre-trained models with advanced LLMs, this study aims to identify the strengths and limitations of each approach in fine-grained opinion detection. This work contributes by advancing the understanding of ***how LLMs outperform or complement existing pre-trained models in complex opinion mining tasks?***.

Background

Sentiment analysis

Sentiment analysis has evolved significantly from identifying simple positive or negative polarity to understanding the complex structure of subjectivity. Initially, sentiment analysis models focused on categorizing text as positive, negative, or neutral, based on basic lexicons or syntactic patterns. While these early models served their purpose, they often struggled to capture the depth of human sentiment, especially when emotions, intentions, and beliefs were involved. Over time, the field shifted towards more fine-grained sentiment analysis, aiming to not just classify sentiment but also analyze its intensity, source, and specific emotion. This evolution is exemplified by the transition from traditional models like Naive Bayes and SVM to more sophisticated deep learning models, which capture richer contextual information [8, 4, 5].

Traditional NLP Models and Their Approach

Traditional NLP models for sentiment analysis, such as Naive Bayes, Support Vector Machines (SVMs), and Recurrent Neural Networks (RNNs), typically rely on handcrafted features, including bag-of-words representations, part-of-speech tagging, and sentiment lexicons. While these methods have demonstrated moderate success, they often struggle with handling the complexity of real-world language. For example, SVMs rely heavily on feature engineering, which is computationally expensive and often limited to the quality of the features used [3]. RNNs provide more flexibility in modeling sequential data but struggle with long-range dependencies [2]. BERT (Bidirectional Encoder Representations from Transformers) and T5 (Text-to-Text Transfer Transformer) are examples of transformer-based models that have significantly advanced the field

of opinion mining use pre-trained embeddings and fine-tuning techniques to capture contextual information, allowing for more accurate classification of sentiment at both the sentence and document levels [1, 7]. However, these models typically focus on binary sentiment classification and fail to address the multi-dimensional aspects of opinion mining, such as opinion type, intensity, and source.

LLMs and Their Application for Opinion Mining

Large Language Models (LLMs) like GPT-3 and BERT-based variants have revolutionized NLP by learning from vast text data to generate contextual embeddings. These models use unsupervised learning to understand complex linguistic patterns and perform various tasks without task-specific supervision [6]. Recent research shows that LLMs excel in sentiment analysis, performing well in simpler tasks and often outshining smaller, domain-specific models in few-shot settings, though challenges persist in structured sentiment phenomena [10]. Techniques for optimizing sentiment analysis using GPT-3 have been proposed, demonstrating that fine-tuning large pre-trained models improves performance and adaptability [9]. Unlike traditional models, LLMs handle complex reasoning, detect opinion sources, recognize sentiment intensity, and capture nuanced emotions in multi-party discourse, making them ideal for fine-grained opinion mining.

Challenges

To design an LLM-based system for scientific discovery, we need to solve the following challenges:

- **Challenge 1:** Identifying Opinion Sources in Complex Sentences
  One of the main difficulties in opinion mining is identifying the source of the opinion - who is expressing the opinion. In many sentences, multiple agents or nested sources may be involved. For instance, the sentence “A thinks B dislikes this idea” contains two agents: A and B. Properly linking the opinion to the correct source is a key challenge in source identification, especially when dealing with ambiguous or complex sentence structures.

- **Challenge 2:** Intensity Detection and Fine-Grained Classification
  While many sentiment analysis models can classify text as positive or negative, they often fail to capture intensity, the degree to which an opinion is expressed. For instance, a statement like “I really like this product” carries a stronger sentiment than “I like this product,” but both would likely be classified as positive. The task of detecting intensity and distinguishing between varying levels of sentiment is a critical challenge in fine-grained opinion mining.

Goals and the Proposed Method

We propose a set of approaches to establish new high baselines for fine-grained opinion mining on the sentiment analysis dataset using pre-trained large language models (LLMs). The main focus is to systematically evaluate how LLMs such as BERT and T5 can be applied to identify multiple dimensions of private states - including type, polarity, intensity, opinion expressions, and sources - and to compare different model architectures, input/output formats, and learning paradigms for these tasks.

Challenge 1: Identifying Opinion Sources in Complex Sentences:

Fine-tuning LLMs to predict opinion attributes (type, polarity, intensity) for individual text spans.

Challenge 2: Intensity Detection and Fine-Grained Classification:

Using T5 to generate structured outputs that capture both opinion expressions and their associated sources, enabling the model to handle complex multi-agent discourse.
The methods will be evaluated by comparing single-task models against multi-task architectures and assessing different input/output formats (e.g., classification vs. generation). These LLM-based models are expected to push the boundaries of multi-dimensional opinion mining, improving the accuracy and flexibility of sentiment analysis in complex texts.

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